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A Eureka Report on:

“Biochemical Engineering For Health And Enviroment”

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Topic : Biosensors For Early Detection of Neurodegenerative Diseases.

Domain : Biochemical Engineering for Health and Environment.

Problem Statement :

Current Neurodegenerative Disease tests are invasive and detect disease too late for early intervention, making ultra-sensitive, non-invasive biosensors crucial for early diagnosis.

Biosensors for Early Detection of Neurodegenerative Diseases

Team: Molecule Mavericks

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Abstract :

Neurodegenerative diseases (NDs) such as Alzheimer's disease and Parkinson's disease are a significant global health burden because they often get diagnosed in late ages when a lot of irreversible neuronal injury has occurred. Existing diagnostic methods are often invasive, expensive, and not sufficiently sensitive for use in the early diagnosis of these diseases. This project bridges this significant gap by suggesting a new non-invasive biosensor that can effectively perform a battery of tests on several crucial biomarkers, in particular proteins.

The approach we will adopt, draws on the founding principles of biochemical engineering and involves creating a next-generation electrochemical biosensor platform. A screen printed electrode will be immobilized by a proprietary nanocomposite material that will boost the signal of detection considerably. The enhanced surface will be functionalized by a proprietary series of bioreceptors (aptamers or antibodies) that will selectively sequester trace biomarkers like amyloid-beta, tau, and alpha-synuclein directly from plasma or saliva.

The main benefit of our approach here is that it can allow presymptomatic diagnosis to be made as soon as possible so that timely therapeutic interventions can be provided. In addition, the biosensor has found a high level of specificity as well as sensitivity coupled with a fast turnaround time without a prohibitively expensive price tag compared to conventional approaches. Its innovation can be attributed here to the working by virtue of the multiplexed detection method coupled with signal amplifier nanomaterials that provide unmatched levels of sensitivity. This portable point of care device holds the promise of being a paradigm shift away from centralized diagnostic protocols towards decentralized screening so that patient care can be improved and the health burden tied up in neurodegenerative disorders can be minimized.

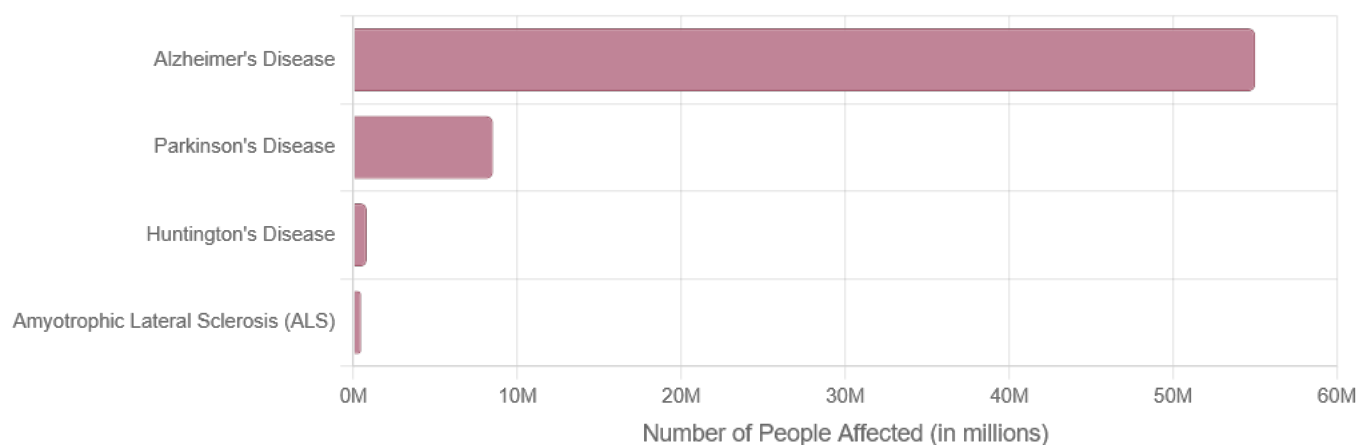
Introduction:

Every year, tens of millions of individuals worldwide are affected by neurodegenerative illnesses, which are defined by the inevitable annihilation of non-specific surrounding brain cells and the physical degradation of target neurons associated with the illness. Neurodegeneration, on the other hand, is defined as the functional loss of neurons and gradual structural, resulting in a wide range of pathological and clinical manifestations, as well as worsening of functional architecture. One of humanity's most pressing issues is reducing the harm caused by neurodegenerative illnesses including Huntington's disease (HD), Lou Gehrig's disease/Amyotrophic Lateral Sclerosis (ALS), Parkinson's disease (PD), and Alzheimer's disease (AD). Early detection of neurodegeneration related disorders that emerge as people age is critical for both the disease's management and the patient's living conditions. The treatment procedure is slowed when illnesses like PD and AD are diagnosed until after symptoms arise. As people live longer, it's more important than ever to understand the pathophysiology of neurodegenerative disease (NDD) so that treatment and diagnostic choices can improve and the costs NDD places on the healthcare system can be reduced. Numerous studies on NDD biomarkers (quantifiable signals indicative of injury, contamination, or illness) have been conducted with the goal of identifying disease occurrences before neurological damage becomes too severe to be reversed by therapeutic drugs. Traditional biomarkers have yet to be established, therefore it's unclear if they can be used to diagnose neurodegenerative illnesses early. More-over, researchers have been forced to collect a sample via peripheral patient-derived biomaterial, that may not accurately reflect the circumstances of neurodegenerative illness due to the difficulties of capturing predominantly disease-associated neurons. Biomarkers, which are unique molecules present in human fluids and are associated with neurodegenerative processes, can improve the early diagnosis of NDD's. Furthermore, micro and nanotechnology advancements have allowed the growth of biosensing devices capable of real-time identification of several biomarkers in therapeutically related samples.

Due to the ability to detect disease-relevant biomarkers in a real-time manner at a cost-effective price with a high degree of sensitive within a short period and label-free, biosensors represent a very promising technology. Biosensors are sensor devices with a built-in receptor transducer that can be utilized to monitor semi-quantitative information or quantifiable. The biosensor is a device with either a piezoelectric, electrochemical, or optical type of signal transducer and can be classified based on the type used. Biosensors have a number of advantages for neurodegenerative disease detection and monitoring. Biosensors provide a stress-free and non-invasive sample collection process, particularly through salivary analysis. The process allows for early detection and early treatment of neurodegenerative diseases. Graphene derivative materials (GDMs) have been incorporated into biosensors with several benefits, including excellent chemical, electrical and mechanical characteristics, good biocompatibility, and surface modifi-

cation via biomolecular recognition. The above features allow for the simple, accurate, and sensitive detection of biomarkers. Electrochemical biosensing has been demonstrated as a cheap solution compared to the standard conventional diagnostic approach with increased sensitivity and specificity. Incorporation of nanomaterials into biosensors also allows for the electron transport in conjunction with enhanced immobilization of the bio-recognition species with a further ability to achieve enhanced biosensor function. Other technologies, such as OCT, also have demonstrated their potential as neurodegenerative disease biomarkers with the assistance of retinal layer thickness as well as providing accurate indices of disease pathology and disease prognosis. Biosensors have a multitude of advantages for the detection of neurodegenerative disease, including non-invasive sampling with high selectivity and sensitivity, as well as a cost-effective price with a potential for early detection. Utilization of sophisticated materials and technologies such as GDMs and OCT allows to optimize rest biosensor potential in this area. The above characteristics support biosensor potential as tools in improving neurodegenerative disease diagnosis, monitoring, and management.

Management and detection of these disorders in their earliest stages are essential for enhancing the quality of life for the elderly. The main goal of identifying preclinical neurodegenerative diseases is to enable the use of treatments during the early stages, which can help delay, reduce, or even prevent the eventual neurological damage that would occur if these issues are not addressed. However, current clinical methods for diagnosing neurodegenerative diseases are often time-consuming and expensive, requiring specialized staff to operate complex machinery. Because of this, there is a growing and urgent need to develop reliable, affordable, convenient, and easy-to-use diagnostic methods. The purpose of this review is to explore neuro-biosensors capable of detecting biomarkers associated with neurodegenerative diseases, as well as to examine the challenges within this field and gain an understanding of the importance of early detection, the role of nano-transducers, biomarkers, various biosensors, and the future direction of this area.



Source:Image generated by Gemini

Chapter 2

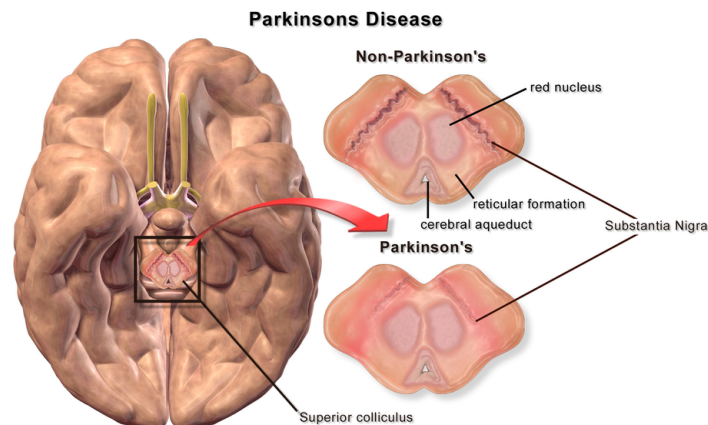
Neurodegenerative Diseases:

Neurodegenerative disorders present themselves through a range of symptoms, such as the selective loss of brain cells, inflammation in the nervous system, the formation of abnormal protein clumps, and damage to neurons and their connections. These issues lead to significant changes in thinking and behavior. Conditions like Alzheimer's disease (AD), Parkinson's disease (PD), amyotrophic lateral sclerosis (ALS), frontotemporal dementia, Huntington's disease (HD), and prion diseases are all types of neurodegenerative disorders that affect a large number of people globally. Among these, PD and AD are the most common neurological diseases. As people get older, the risk of developing a neurodegenerative disorder increases significantly. Recent studies suggest that a large number of people will be affected by neurodegenerative conditions in the coming decades, highlighting the urgent need to understand the causes and develop new methods for managing and treating these disorders. Research estimates that approximately 30 million people around the world currently live with Alzheimer's disease. The health, economic, and social impacts of neurodegenerative disorders are substantial. However, the exact mechanisms behind these disorders are still not fully understood, and there are currently limited effective treatments available. Many factors contribute to the difficulty in managing and treating these conditions, including the presence of the blood-brain barrier (BBB), which acts as a protective layer that prevents most drugs from reaching the brain and thus only treats the symptoms rather than the underlying disease. Additionally, the diagnostic markers used for these conditions are not well established and may not accurately reflect the presence of possible or likely diseases. Neurological disorders are increasingly contributing to a large number of disability-adjusted life years, especially in high-income countries.[1-2]

2.1 Parkinson's Disease :

Parkinson's disease, commonly referred to as Parkinson's, represents a form of neurodegenerative disorder primarily impacting the central nervous system. This condition affects both motor and non-motor capabilities. The motor manifestations, collectively known as parkinsonism, encompass symptoms such as tremors, difficulties in initiating movement, rigidity of muscles, and challenges in sustaining balance. Concurrently, non-motor manifestations may present issues related to the autonomic nervous system, disturbances in sleep patterns, a diminished sense of smell, and alterations in behavior or mental health, including cognitive deterioration, psychosis, and anxiety, which can emerge at any stage of the disease. These manifestations typically evolve gradually, and as the condition advances, the non-motor symptoms often become increasingly pronounced.[3] A majority of Parkinson's disease instances are classified as idiopathic, indicating that the precise eti-

ology remains unidentified, despite the recognition of certain contributing factors. The condition arises from the progressive degeneration of neuronal cells within the substantia nigra, a brain region responsible for dopamine production, a neurotransmitter essential for regulating movement. Although the precise mechanisms underlying this neuronal loss are not entirely elucidated, it is associated with the development of atypical protein aggregates known as Lewy bodies within the brain cells. Additional potential contributors include genetic susceptibilities, environmental factors, pharmacological influences, lifestyle decisions, and prior health conditions. Physicians generally diagnose Parkinson's disease by evaluating the clinical signs and symptoms noted during a neurological assessment, particularly those pertinent to movement. Furthermore, advanced imaging techniques, such as positron emission tomography, may be employed to assist in the diagnostic process. Parkinson's disease typically manifests in individuals aged over 60, with approximately one percent of this demographic being affected. When the onset occurs in individuals younger than 50 years of age, it is designated as early-onset Parkinson's disease.[4]

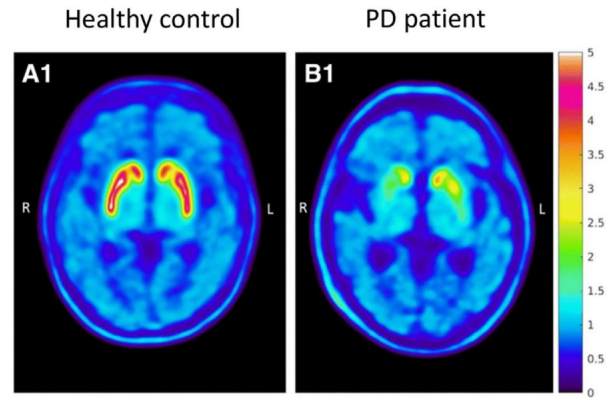


Source: <https://en.wikipedia.org/wiki/Parkinsons>

Parkinson's results from the death of dopamine-releasing neurons, seen by the loss of dark neuromelanin in the lower inset.

Reduced radio isotopic uptake in the striatum of a PD patient, captured through PET.

PD is mostly associated with death or damage of the neurons or nerve cells in the brain that govern body movement. This leads to decreased DA release, which ultimately results in problems with movement. PD patients also lose nerve endings that generate norepinephrine. There are no particular treatments for PD at the moment, and the therapies, surgeries, and medications provided can only alleviate the illness's symptoms. Among the therapy options for PD, the usage of proteins like human glial cell line-derived neurotrophic factor (hGDNF) has shown promise. Ansorena et al. developed a straight-

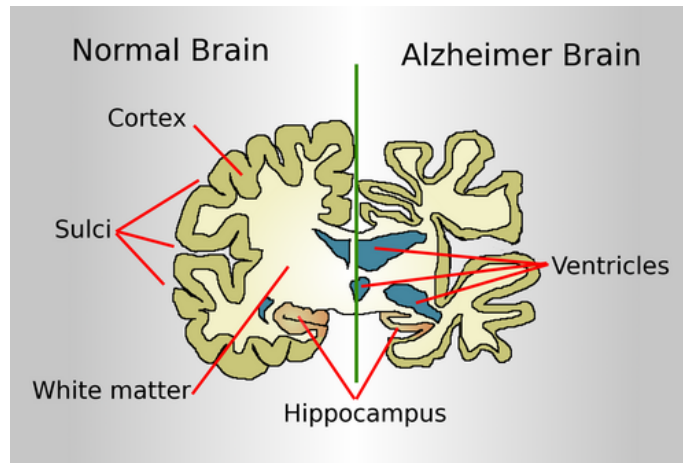


Source: <https://en.wikipedia.org/wiki/Parkinsons>

forward and rapid approach for producing a large concentration of pure hGDNF utilizing a mammal cell- derived technology. In addition, PD has been detected via NP-mediated biomarker detection (Table 1). PD is linked with a specific degeneration of DA substantia nigra and a drop in the striatum's DA level. This deficiency of DA results in motor symptoms (resting tremors, stiffness, and so on) as well as other symptoms such as cognitive impairment . PD, the 2ndmost common NDD, is likewise marked by α -synuclein amyloid build-up. Adam et al. developed a biosensor capable of detecting a particular PD biomarker, such as α -synuclein aggregation, which is critical for minimizing the burden of PD and for early detection. Detecting alpha-uneven synuclein aggregation is a potential tool for the early identification of PD. The ZnO nanocomposite affixed to the aluminium microelectrode surface offers an advantageous substrate for effective antibody loading through antigen α -synuclein binding. Aghili et al., on the other hand, created an electrochemical nanobiosensor to detect PD early using the measurement of a circulating biomarker, miR-195. Gold nanowires (GNWs) and exfoliated graphene oxide were used to enhance the surface of a screen-printed carbon electrode (EGO). A single-strand thiolated reagent was designed to precisely hybridize with the target miRNA (miR-195), and doxorubicin was used as an electrochemical indicator for differential pulse voltammetry studies. Based on the findings, physicians may consider using the miR-195 electrochemical nanobiosensor for the medical diagnosis of PD.[4-8]

2.2 Alzheimer's Disease :

Alzheimer's disease is a type of neurodegenerative illness and is the most common cause of dementia, affecting about 60 to 70 percent of people with dementia. The earliest sign is usually trouble remembering recent events. As the condition progresses, more problems appear, such as difficulty with language, confusion (including getting lost easily), changes in mood, lack of interest, poor care for oneself, and behavioural changes. As the illness gets worse, people often become less interested in interacting with family and



Source: <https://en.wikipedia.org/wiki/Alzheimer>

Figure 1: Alzheimer Diseases

friends. Over time, they may lose control of basic bodily functions, eventually leading to death. While the rate at which the disease progresses can differ from person to person, the average time from diagnosis to death is between three and twelve years. The exact causes of Alzheimer's disease are not fully understood, but there are several environmental and genetic factors that may increase the risk. The strongest known genetic risk factor is a particular version of the apolipoprotein E gene. Other possible risk factors include a history of head injuries, depression, and high blood pressure. The disease typically progresses through the build-up of abnormal proteins in the brain's outer layer, known as amyloid plaques and neurofibrillary tangles. These abnormal proteins disrupt normal brain function and, over time, cause permanent damage to brain cells and the connections between them. A likely diagnosis is made based on the patient's medical history and cognitive tests, along with imaging and blood tests to eliminate other possible causes. Early symptoms are often confused with the natural changes that happen in aging brains. A definitive diagnosis can only be made by examining brain tissue, which is only possible after the person has passed away.[6-9]

AD has a scientific history dating back to 1906 when Dr. Alois Alzheimer examined the brain of a lady who died prematurely due to a mental disorder. In the brain tissues, he discovered several twisted bundles of fibers and aberrant clusters. These tangles and plaques are thought to be the disease's primary hallmark. AD is most often associated with memory difficulties, although additional symptoms include vision impairments, judgment or poor thinking, trouble locating words, and others. As the condition continues, behavioural and personality changes, difficulty doing everyday duties, being lost and wandering, and difficulty paying and managing cash occur and memory loss becomes more severe. Further phases result in the patient's inability, increased bewilderment, and

loss of linguistic control to recognize friends and family. Eventually, the plaques and tangles expand throughout the brain, leaving the sufferer utterly reliant on others. AD therapy is very difficult since no one medicine can adequately cure the illness. At the moment, the definitive diagnosis of AD is achievable only upon postmortem neuropathological investigation. Available diagnostics for suspected patients are prohibitively expensive, limited in availability, or invasive. There is still a need for early detection technologies to improve existing therapy, even before mild cognitive impairment occurs (MCI). The deposition of $A\beta$ in extracellular amyloid plaques is one of the hallmarks of AD, resulting in severe neurodegeneration on a local level. Visualization of these amyloid plaques in living tissues is crucial for determining therapy success, monitoring disease development, and diagnosing AD. Another AD-specific lesion occurs as a consequence of hyperphosphorylation of Tau protein, which causes the protein to dissociate from microtubules and aggregate form neurofibrillary tangles in the intracellular environment. As a consequence, substantial interest has been shown in creating a molecular imaging agent capable of detecting AD lesions based on these clinical characteristics (Table 2). These agents must be nontoxic and, preferably, capable of crossing the BBB without the need for facilitation. Due to reproducibility and stability issues in biological samples, there are many substantial hurdles associated with the scalable and regular manufacture of biosensing devices. Additionally, relatively few studies employed genuine samples of AD patients, and the majority of biosensors were evaluated in the buffer, manufactured samples, or real specimens spiked with the desired analytes. Because AD illness is caused by several pathogenic pathways, detecting multipathing is a critical step that is currently sadly lacking. As a result, the panel detection of biomarkers is crucial for enabling reliable and sensitive detection if one is required. Additionally, there are the following difficulties :

- (a) A mass new knowledge about the pathogenesis of AD and identify better and novel biomarkers.
- (b) Advances in the development of a repeatable biomarker approach.
- (c) To minimize matrix interferences, pre-treatment samples, such as extraction and purification, are necessary.
- (d) Biosensor miniaturization and integration into a single application platform.
- (e) Cost savings associated with developing and implementing methods NMs are critical in resolving some of the aforementioned difficulties.

Numerous NMs of varying sizes display unique properties that manifest themselves in a variety of optical and electrochemical activities. When biological recognition components and NMs are coupled, an improved diagnostic method emerges. Kang et al. describe the development of a new poly-L-lysine (PLL)-mediated nanobiosensor for the invitro detection of Amyloid. The PLL molecules were used as an $A\beta$

detection signal amplifier. Amyloid has been detected using both the indirect enzyme-linked immunosorbent test (ELISA) and the sandwich ELISA methods. PLL was used to modify a commercially available ELISA plate, and the amplified amino groups were triggered using a functional group specific for Amyloid binding. As a consequence, the PLL-mediated indirect ELISA nanobiosensor demonstrated the highest sensitivity for Amyloid detection. Alternatively, Amini et al. suggested an improved nanobiosensor for AD detection in their study. They improved the metal layer thickness for copper, aluminum, silver, and gold metals using a modified approach to enhance the quality and sensitivity factor. An optimal result is achieved by using gold as the active layer and incoming light with a wavelength and angle of 632 nm and 46, respectively. Finally, AD detection is computed utilizing the improved surface plasmon resonance (SPR) structure. The researchers can measure even minute amounts of molecular structure present in blood samples and cerebrospinal fluid (CSF) using this approach.[10-12]

Marker of disorder	NP	Diagnosis modality	Experiment type	BBB crossing	Ref.
α -Synuclein	Immuno/magnetic particle	Immunoassay	In vitro	Not applicable	[59]
	Gold nanorod	Surface plasmon	In vitro	Not applicable	[60]
DA receptor	Immuno-targeted far-red QDs	Fluorescence	In vivo (acute rat brain slide)	Intraventricular injection	[61]
DA	PEGylated PFPBA NPs (100 nm)	Near-infrared fluorescence	In vivo (mouse)	Yes	[62]
	PEGylated PFPBA NPs (120 nm)	Fluorescence quenching	In vivo (zebrafish)	Yes	[63]

Source: <https://www.researchgate.net/publication/359911768>

Figure 2: Table 1

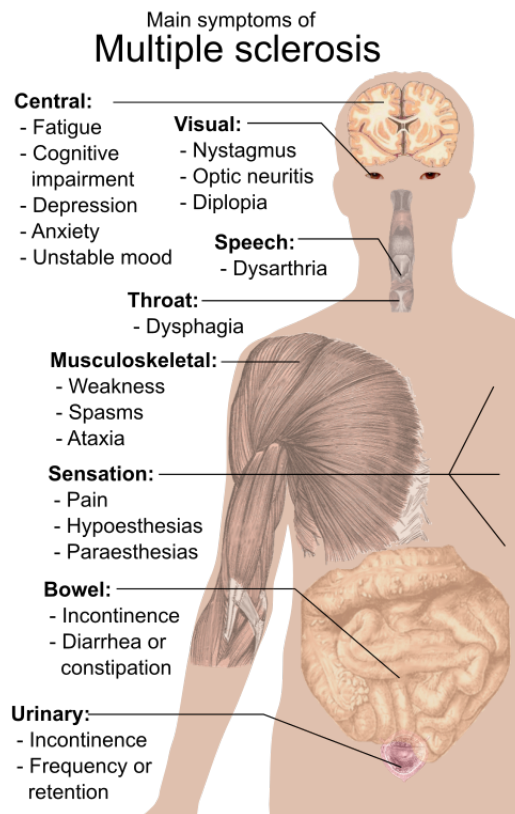
2.3 Multiple Sclerosis :

Multiple sclerosis (MS) is an autoimmune condition that causes damage to the myelin sheath, which is the protective covering of nerve cells in the brain and spinal cord. As a demyelinating disease, MS interferes with the nervous system's ability to send signals, leading to various physical, mental, and in some cases psychiatric symptoms. Common signs and symptoms include double vision, loss of vision, eye pain, muscle weakness, and a loss of sensation or coordination.

MS presents in several forms : One form involves new symptoms appearing suddenly and then improving, known as relapsing-remitting MS, which is the most common type,

affecting around 85% gradually get worse over time, a condition referred to as primary progressive MS, which is found in about 15% of individuals with relapsing-remitting MS may experience a slow worsening of symptoms after attacks, which is called secondary progressive MS. In relapsing forms of MS, symptoms may go away completely between attacks, although some neurological issues may remain, especially as the disease progresses. In progressive forms of MS, the body's function gradually declines once symptoms start, and the condition tends to worsen if not treated. A person may experience a single episode of neurological symptoms without meeting all the criteria for an MS diagnosis. This is called a clinically isolated syndrome. The exact cause of MS is not known, but it is believed to involve either the immune system attacking or the inactivation of cells that produce myelin. Possible contributing factors include immune system dysfunction, genetic predisposition, and environmental influences, such as viral infections. The McDonald criteria are a set of updated guidelines used to help diagnose MS.

There is currently no cure for MS. However, treatments are available to reduce inflammation and symptoms during flare-ups, prevent future attacks, and slow the progression of the disease, ultimately helping to improve the patient's quality of life.[5-6]



Source: <https://en.wikipedia.org/wiki/Multiplesclerosis>

Recent breakthroughs in the area of nanoscience provide a novel approach to overcom-

ing these obstacles and establishing a more fertile ground for monitoring neurological illnesses and innovation in medication development. Numerous neuroimaging methods are available to assist in diagnosing the neurological complications associated with neurodegenerative illnesses. Single photon emission CT, positron emission tomography, and nuclear medicine research are all examples of these approaches. Magnetic resonance spectroscopic imaging, ^1H , and ^{31}P magnetic resonance spectroscopy, and CT and MRI are all morpho-logical investigations. Nanotechnology's recent advancements are critical in the biomedical industry because they provide enhance delectrophysiological and neuroimaging techniques for fundamental clinical research. Nanotechnology aids in the progression and discovery of central nervous system (CNS) illnesses, providing the creation of innovative diagnostic methods and fresh insight into CNS physiology. Nanotechnology-based systems are utilized to increase the power of neuroimaging by using more precisely targeted probes of molecular imaging and contrast. In addition, it may be integrated into a sophisticated biosensor system inside the brain for testing circuit physiological principles. Using the unique and improved biological, chemical, and physical features of NMs, may be considerably improved present approaches for detecting severe CNS illnesses and used new insights into brain physiology to generate innovative treatment options. These strategies may be used to deduce the underlying neurological mechanisms that contribute to illness development. Thus, diffusion tensor imaging (DTI), CT, and MRI are often utilized in conjunction with neurological and clinical evaluations to aid in the identification of many degenerative CNS illnesses. Due to the heterogeneity of MS, which is characterized by diverse demyelination patterns, it is exceedingly implausible that a single diagnostic marker would cover the whole range of MS subtypes. Lolli et al. created CSF114 (Glc), a synthetic glycoprotein antigen probe, for the detection of autoantibodies seen in the blood of MS patients. The authors demonstrated that CSF114(Glc) antibodies identified oligodendrocyte and myelin autoantigens in human brain tissue. This understanding enables the creation of a unique approach for identifying MS patients who had demyelination caused by antibodies, a subgroup of MS patients. Later that year, the same group published a paper describing the creation of a gold SPR biosensor with covalently bonded CSF114(Glc) for real-time detection of multiple sclerosis (MS) in blood. When utilized to differentiate MS patients from healthy blood donors, this SPR biosensor demonstrated a poor sensitivity (36%), but a high specificity (95%). Apart from MS detection, further clinical correlation and multiple autoantibody identification may be utilized to guide treatment and track its response. MS also has been detected via NP-mediated biomarker diagnosis.[13-18]

Marker of disorder	NP	Diagnosis modality	Experiment type	BBB crossing
Tau tangles	Gold NP-anti-tau	Two-photon Rayleigh scattering	In-vitro	No need of BBB Crossing (CSF)
Cerebrovascular amyloid deposit	Cross-linked chitosan	single-photon emission computed tomography (CT)/ Magnetic resonance imaging (MRI)	In-vivo	Antibody targeted to the vessel wall
	Monocrystalline iron oxide NPs	MRI	Ex-vivo	No need of BBB crossing (vessel's wall)
Amyloid plaques	NPs- Bovine Serum Albumin-Sialic Acid	MRI	In-vivo	Without any facilitation
		Fluorescence	In-vitro	
	Curcumin-magnetic NP	Immunohistochemistry	In-vitro	Without any facilitation
	Liposomes-ET6-21	MRI	In-vivo	Without any facilitation
		Immunohistochemistry	Near-infrared	
	Ultrasmall particles of iron oxide-PEG-A β (1-42)	μ MRI	Ex-vivo	Without any facilitation
			In-vivo	
	Ultrasmall particles of iron oxide-PHO	MRI	In-vivo	Without any facilitation
	Magnetic iron oxide NPs-antiferitin	MRI	Ex-vivo	Facilitated (mannitol)
		Immunofluorescence	In-vitro	

Figure 3: Table 2

Source: <https://www.researchgate.net/publication/359911768>

2.4 Amyotrophic lateral sclerosis :

ALS, also known as Lou Gehrig's disease or motor neuron disease (MND), is a rapidly progressing neurodegenerative disorder that harms the neurons responsible for controlling muscle movement. Globally, the incidence of ALS is approximately one to three cases per 100,000 people. Currently, there is no effective treatment for ALS, and the average survival time after diagnosis is between 24 and 48 months. ALS mainly affects motor neurons in the spinal cord, brain, and brainstem, leading to progressive muscle wasting and degeneration of motor neurons, which ultimately results in paralysis and death due to failure of the respiratory system. There have been developments in identifying potential biomarkers for both sporadic and inherited forms of ALS. As mentioned earlier, the Bcl-2-SODox complex subtype found in individuals with sporadic and familial ALS could serve as a biomarker due to the similar structure of the mutant SOD1 protein that binds to mitochondrial Bcl-2. The predictive ability of cystatin c in patient cerebrospinal fluid (CSF) was evaluated using a quantitative ELISA method. Although cystatin c levels are lower in ALS patients compared to healthy individuals and the ELISA test correlates

with disease progression and patient survival, it was found to not be a reliable predictor of ALS. However, a three-protein CSF biomarker panel identified by Pasinetti et al., which includes the peptic fragment of the neurosecretory protein VGF and cystatin c, showed the best prediction of ALS with 91 percent sensitivity and 97 percent specificity when compared to using individual proteins alone. In individuals with ALS, there was a decrease in ALS-specific proteins compared to healthy controls. Another panel of 14 biomarkers was discovered, consisting of analytes related to iron balance, growth factors, and pro- and anti-inflammatory cytokines. Five of these proteins accurately distinguished ALS patients from healthy controls with an accuracy of 89.2 percent, a sensitivity of 87.5 percent, and a specificity of 91.2 percent.[6,14,19,20]

Marker of disorder	NP	Diagnosis modality	Experiment type	BBB cross-ing
Peripheral immune cell infiltration	Iron oxide NPs in T cells	MRI	In vivo (mouse)	Yes
	Iron oxide NPs in T cells	MRI	In vivo (mouse)	Yes
Immune cells activation	Iron oxide NPs	MRI	In vivo (mouse model of MS)	Yes
Vascular inflammation	An-ti-VCAM-1 magnetic particle iron oxide	MRI	In vivo (mouse model of MS)	Yes
	Anti-ICAM-1 magnetic particle iron oxide	MRI	In vivo (mouse model of MS)	Yes

Figure 4: Table 3

Source: <https://www.researchgate.net/publication/359911768>

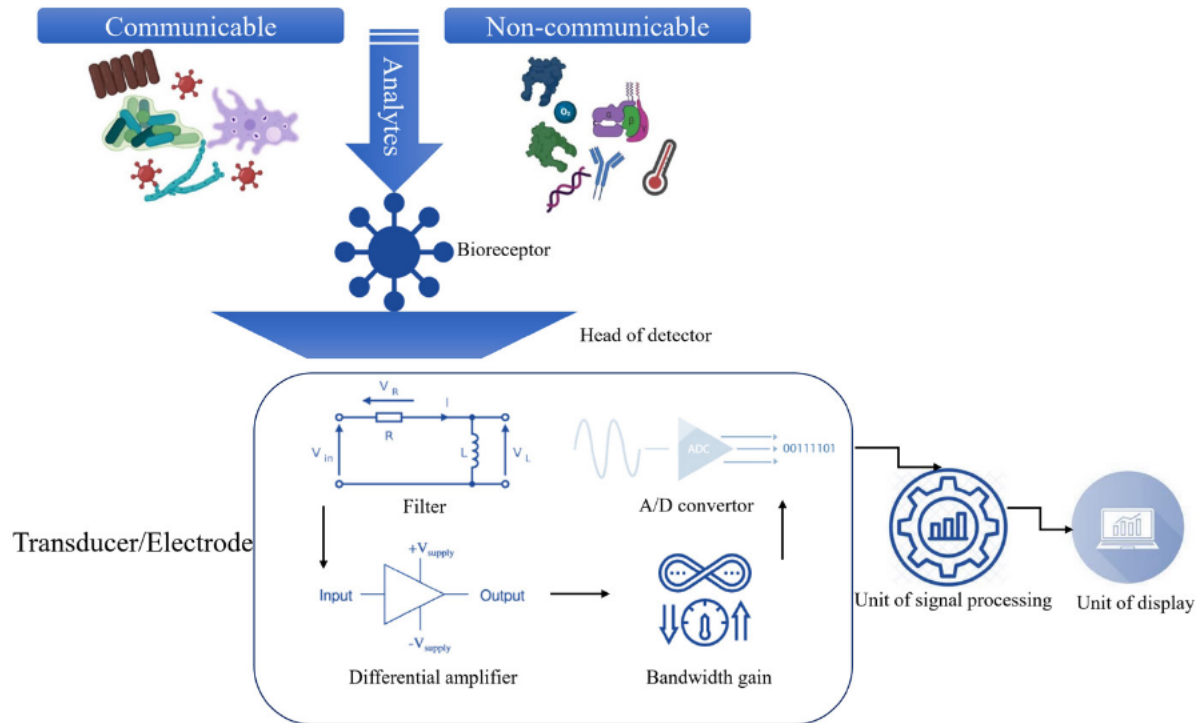


Figure 5: A working biosensor's block prototype
Source: <https://www.researchgate.net/publication/359911768>

Chapter 3

Importance Of Early Diagnosis:

Neurodegenerative illnesses are detected years after they begin, when the majority of particular neurons die and the brain's neuroplasticity is depleted, by the manifestation of specific symptoms. This explains why conventional symptomatic treatment is ineffective. It is considered that the development of preventative treatment and early (preclinical) diagnosis may extend a patient's pleasant life during the preclinical period. Despite significant attempts, no early diagnosis for neurodegenerative disorders has been created. This calls into question the process utilized to produce it. The term "neurological disorder" refers to any conditions produced by nervous system or brain malfunction that manifest as psychological and/or physical symptoms. Neurological illnesses are the second cause of disability worldwide, accounting for 276 million disability-adjusted life years(DALYs) annually, with 80 percent occurring in middle- and low-income countries. No one examination can provide a conclusive diagnosis for the majority of neurological diseases. Numerous neurological tests are performed in conjunction with EEG (electroencephalography), EMG (electromyography), and MRI. Enzymatic evaluations (for

example, hexosaminidase A Tay–Sachs assay) and immunosorbent tests (for example, ELISA for Alzheimer’s amyloid-peptides) are both used in conventional biochemical research. Standard genetic testing is used to decide whether to utilize polymerase chain reaction (PCR); for instance, allele-specific Tay–Sachs PCR or RT-PCR (for example, retroviral detection). Traditional immunoassays have limitations in terms of scientific automation, time, and accuracy. Recovery from any illness and treatment is essentially determined by the efficacy of the diagnostic procedures and their early-phase detention. Even though immunofluorescence (FRS), immunosorbent approaches, and microscopic method shave been proved to be therapeutically essential in a variety of disorders treatment. They do, however, have several drawbacks, including their bulky nature, cost, inaccuracy, stumpy specificity, and reduced sensitivity. To address these problems, high-throughput, effective, biocompatible, and quick analytical techniques are becoming more necessary in healthcare/clinical/biomedical. Even with advances in scientific understanding, mankind continues to face several issues related to non-communicable and communicable illnesses. As indicated in the Introduction, the most effective strategies for detaining and preventing such illnesses, as well as the efficacy of treatment and diagnostic processes, are early detection and prevention, as well as the effectiveness of treatment and diagnosis procedures. As a result, various novel techniques, such as the use of biosensors/nanobiosensors for illness detection and therapy, have continued to be beneficial in this area.[1-14]

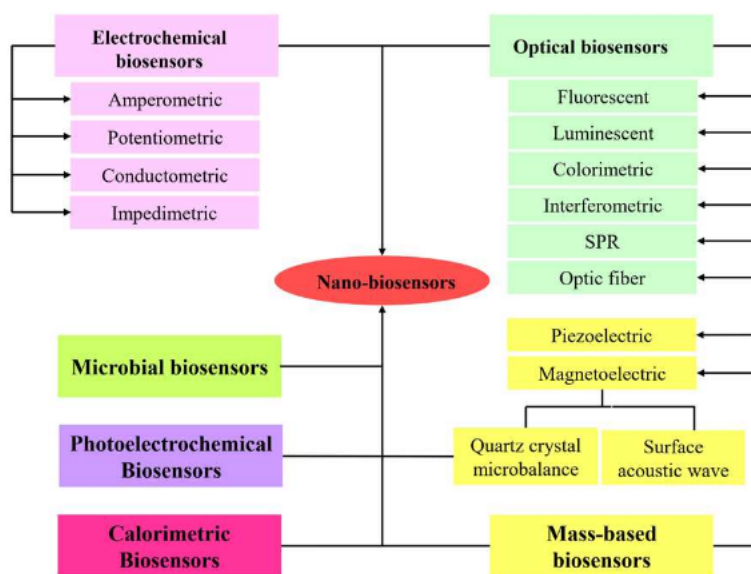


Figure 6: Most prevalent biosensors types and subtypes
 Source: <https://www.researchgate.net/publication/359911768>

Chapter 4

Importance Of Nano Transducers:

Whether the condition is contagious or not, recovery and treatment are dependent on a timely and accurate diagnosis. While immunofluorescence, immunosorbent assays (ELISA), and microscopic techniques are clinically significant, they all have certain disadvantages, including high cost, inaccurate findings, low specificity, inconvenient nature, and limited sensitivity. To overcome the aforementioned disadvantages, clinical needs for high-throughput, biocompatible, effective, and rapid diagnostic procedures remain unfulfilled. Various sensing approaches have been used in recent decades to detect and identify biological markers linked with NCD and communicable diseases. These technologies use high-conductivity electrodes capable of tracking or detecting electro-active proteins present in the body that are specific to an illness condition and provide a strong signal. All of the above-mentioned features are present in a sensing device referred to as a biosensor. Thus, using biosensors to detect biological markers may be a viable strategy. Additionally, recent breakthroughs in biosensor technology have ushered in evolutionary changes in a variety of sectors, including food processing, agriculture, healthcare, and biological research. At the moment, biosensors are generally classified according to their biological element, which may be a nucleic acid or enzyme, antibody, or their transducing element, which may be calorimetric, optical, acoustic, or electrochemical. The enormous surface area of NPs acts as a powerful transducer, and so combining the NMs with electrical systems might result in proactive electrical transduction processes in smart nano-electromechanical devices (NEMS). Without a doubt, contemporary biosensor mechanisms have several advantages, including observable, quick, and increased sensitivity and accuracy, and multiplicative results when compared to previous glucose/chemical-based biosensors. Biosensors use tissue-specific macro-molecules, organic organelles, microorganisms, enzymes, and immuno-sensors (antibodies) as detecting/sensing mechanisms. The methods of transduction are based on the physio-chemical distinction generated by detecting/sensing processes. As a result, several biosensor transducer mechanisms include calorimetric, optical, piezoelectric, and electro-chemical. Acoustic and ultrasonic are two of the most significant biosensor methods based on piezoelectric transducers; electrochemical, amperometric, and conductometric are three of the most important electrochemical transducer biosensor mechanisms; and optical transducer biosensor mechanisms include fluorescence, absorbance, and chemiluminescence (CL). Biosensors work on the concept of cell signaling, and as previously said, the primary components of a functioning biosensor are electronic parameters, a bio-transducer, and a biorecognition element that may include a monitor/display, amplifier, and a microprocessor. The sensing or biorecognition component is a bioreceptor that is specifically intended to act or detect on a certain analyte (the target substance) whose related qualities are to be assessed or

recognized. The transducer gets input from the bioreceptor and transmits a signal to the signal processor, depending on the transducer used. The amplitude of the output signal is proportional to the concentration of the analyte. Following that, electrical equipment amplifies and processes the signal. In the case of an amperometric sensor, the bioreceptor consists of a specialized bio-material that is kept in close proximity to the transducer or deactivated enzyme. The analyte has a chemical reaction with the bio-material. This results in the formation of a new analyte that provides the calculable electrical response. Occasionally, an analyte is transported to the system which is discharged, cooled, or heated with hydrogen ions or electrons. The transducer could then control the associated mechanism and convert it to electrical signals that could be calculated and adjusted. A biosensor's components are critical to its functioning. To begin, each biosensor is designed to accomplish a certain purpose. The mode of operation of a biosensor is determined by the bioreceptor (antibody, phage, DNA, enzyme, etc.) and the sensing technique (the functions of the transducer). The electrical signal from the transducer is usually tiny and is superimposed over a pretty high baseline. Typically, signal processing begins with the derivation of a signal indicating the baseline position from a comparable transducer that is not coated with any bio-catalyst. The relatively sluggish response rate of the biosensor considerably alleviates the problem of electrical noise filtering. At this phase, the direct output is an analog signal. On the other hand, the signal maybe digitized and sent to a microprocessor unit, which analyses the data, routes it via selected units, and exports it to a data storage. A biosensor may operate within or outside of a live creature; nonetheless, the characteristic parameter being measured/detected by the instrument is often derived from the environment or organism. Fig. 1 illustrates the primary components of a working biosensor with display and processing capabilities. As a consequence, Fig. 2 depicts many different kinds of biosensors, including mass-based biosensors, calorimetric biosensors, photoelectrochemical biosensors, microbiological biosensors, and optical biosensors, and electrochemical biosensors, as well as their subtypes. Electrochemical biosensors are a subclass of biosensors that incorporate an electrochemical transducer and are classified as impedimetric, conductometric, potentiometric, and amperometric sensors, whereas optical biosensors incorporate an optical transducer system and a bio-recognition sensing constituent. Optical biosensors are classified into the following categories: optic fiber, SPR, interferometric, colorimetric, luminescent, and fluorescent. A microbial biosensor, on the other hand, is a device that uses a biomolecule (a component of a microbe) as a transducer to provide a measurable signal indicative of the analyte concentration. Photo-electrochemical biosensors utilize the process where light is converted into electrical energy, which happens at the same time as light absorption and charge separation. Calorimetric biosensors measure the heat absorbed or released during a chemical reaction. In general, mass-based biosensors detect binding events by identifying the mass increase on the sensor surface, which is indicated by a change in the oscillation of the surface acoustic wave.[6-14]

Chapter 5

Biosensors:

5.1 Advantages of Biosensors :

Biosensors have several benefits for the detection and monitoring of neuro-degenerative diseases : Biosensors offer a stress-free and non- invasive means of sample collection, especially via salivary analysis. This method enables early diagnosis and prompt treatment of neurodegenerative diseases. Graphene derivative materials (GDMs) have been used in biosensors with great advantages, such as superior chemical, electrical, and mechanical properties, good biocompatibility, and surface modification for biomolecular recognition. These characteristics facilitate easy, precise, and sensitive detection of biomarkers. Electrochemical biosensing has shown to be a low-cost solution compared to the conventional diagnostic method, with enhanced sensitivity and specificity. The introduction of nano-materials within biosensors also facilitates electron transport and enhances immobilization of bio- recognition species, further allowing for better biosensor performance. Other technologies, like optical coherence tomography (OCT), also have shown their potential as neurodegenerative disease biomarkers using retinal layer thickness and providing precise indicators of disease pathology and prognosis. Biosensors possess many benefits for the detection of neurodegenerative disease, such as non-invasive sampling, high selectivity and sensitivity, cost-effectiveness, and the possibility of early detection. The use of advanced materials and technologies like GDMs and OCT maximizes the potential of biosensors in this aspect. These features contribute to making biosensors a potential aid in enhancing the diagnosis, monitoring, and management of neurodegenerative diseases.[14-16]

5.2 Example of Biosensor Technology :

Biosensor technology includes various devices designed to detect specific substances by combining biological components that recognize these substances with elements that convert the recognition into a measurable signal. These devices are known for their high sensitivity and ability to distinguish between different substances, making them essential tools in medical testing, environmental monitoring, and ensuring food safety. In healthcare, biosensors enable fast disease detection and continuous health tracking, while in environmental and food sectors, they help identify harmful substances and ensure product quality. Their accuracy and adaptability continue to drive innovation across many scientific and industrial areas. Electrochemical biosensors are the most widely used and commercially available type of biosensor. Enzyme-based electrochemical methods are

used in glucose meters, which are a successful example of biosensor technology. Electrochemical biosensors are also used to detect nucleic acids, proteins, and other metabolites, although these are not as commonly available as glucose meters. Notably, while electrochemical biosensors dominate the market, optical biosensors, especially those based on surface plasmon resonance (SPR), have made significant progress. Fiber optic SPR biosensors offer label-free detection of chemical and biological materials using different probe designs like U-shaped bends and tapered structures. Gold nanoparticles are frequently used in optical and non-optical biosensors, including piezoelectric and electrochemical types, due to their unique properties. Biosensor technology continues to evolve, with new developments focusing on nanotechnology, miniaturized lab-on-a-chip devices, and point-of-care testing. DNA-based methods are being integrated into immuno-sensors to improve their sensitivity and address existing challenges. The development of portable and easy-to-use devices, such as screen-printed electrodes with miniaturized sensors, is making biosensors more accessible for use in the field. As research progresses, we can expect a wider variety of biosensing devices to become available for everyday consumer use soon. [6,14,24]

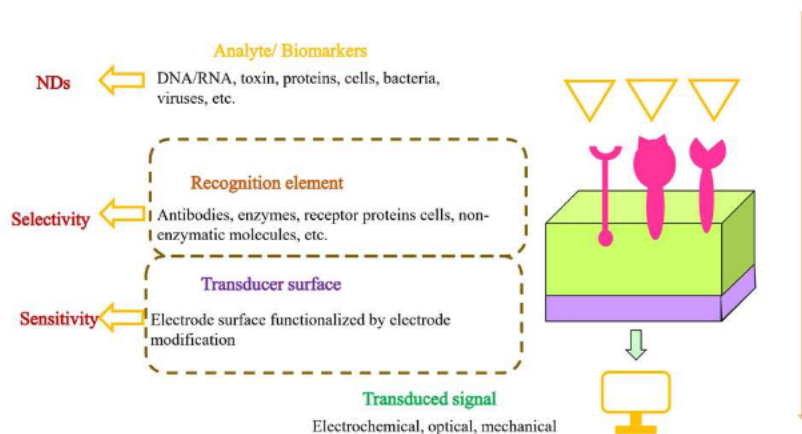


Figure 7: The basic concept of biosensor system and the major components used to detect NDs
Source: <https://www.researchgate.net/publication/359911768>

5.2.1 Optical Biosensors :

Specificity and sensitivity are two critical parameters to consider when evaluating a biosensor. A basic biosensor consists of a biological recognition section capable of capturing the desired analyte and the sensing section capable of converting the biological variation to chemical or physical signals. Specificity is characterized by the available combination of biological sensing sections and recognition, whereas sen-

sitivity is determined primarily by the receptor, which may be a microbe, enzyme, nucleic acid (e.g. aptamer), or antibody, as well as the immobilization sensing, substrate, and process approach. In recent years, optical substrates constructed from NMs with unique optical characteristics such as quantum dots, carbon nanotubes, graphene, and Ag and Au NPs have been extensively exploited to boost the sensitivity of target detection. For example, quantum dots are regarded as a fluorophore for fluorescence detection due to their broad absorption band and photobleaching resistance. Metallic NPs are utilized to improve the signal in SERS-based detection. In general, several kinds of NMs perform a variety of functions in a biosensor-based system. The development of optical biosensors for target detection based on NMs is a novel and current trend in the area of analytical diagnostics. Due to its rapid detection time, simplicity of use, and high sensitivity, optical biosensors based on NMs have emerged as a potentially useful analytical tool for cancer diagnostics. The optical techniques for exosomal cancer biomarkers are classified according to the optical detection methods used, which include fluorescence, electrochemiluminescence (ECL), CL, immunochromatographic assay (ICA), colorimetric, SPR, and SERS (Fig.3). The work by Shawky et al. set out to develop a simple apparatus for the absolute measurement and detection of nucleic acid transcripts by using an optical biosensor based on gold NPs. The nucleic acid transcripts tyrosyl DNA phosphodiesterase 2 (TDP2) and topoisomerase 1 (TOP1) were selected as markers of genomic instability due to their association with a number of neurologic and malignant disorders. The obtained mRNA was promptly measured and identified using the gold aggregating gold (GAG) test, which eliminates the need for amplification, which is presently necessary for transcript quantification. Instead of difficult, expensive, and time-consuming real-time PCR, the GAG test may be used to determine the absolute amount of RNA in many applications. Haes et al. developed a nanoscale optical biosensor based on localized SPR spectroscopy to monitor anti-ADDL specific antibodies, amyloid- derived diffusible ligands (ADDLs), and antigen contact. The study of human brain extracts and cerebrospinal fluid samples from healthy controls and AD patients indicates that the LSPR nanosensor gives fresh information useful for the understanding and potential diagnosis of AD.[6,14,21,22,24]

Elements of a Biosensor

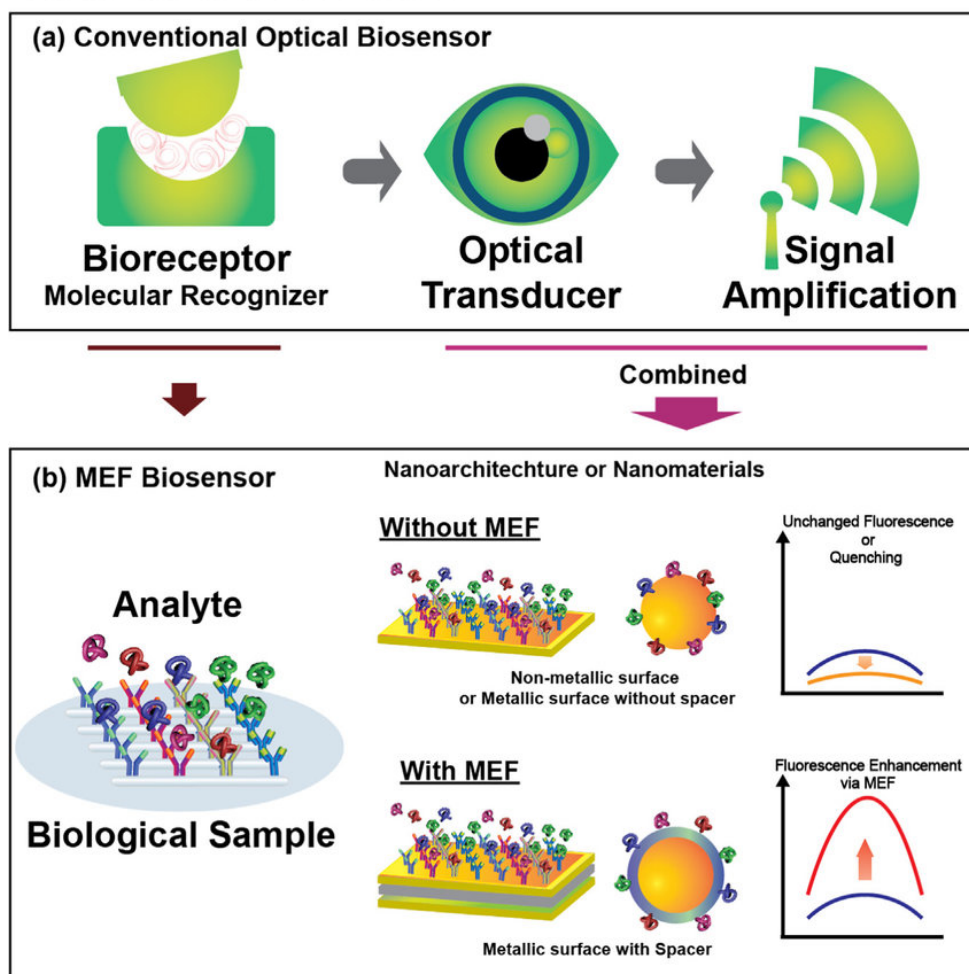


Figure 8: Concept of Optical Biosensors

Source: <https://www.researchgate.net/figure/The-concept-of-optical-biosensors-a-Conventional-optical-biosensor>

5.2.2 Electrochemical Biosensors :

The electrochemical sensor's essential principle is the process by which a variable or constant voltage is supplied to the electrode, and the detection material changes on the electrode surface, resulting in the formation of an electrical signal. Electrochemical sensors are primarily used to determine the electrochemical and electrical characteristics of target substances or molecules in order to perform quantitative or qualitative detection and analysis. Chemically modified electrodes are a popular topic of study in the electrochemical area at the moment. At the moment, the majority of research is concentrated on electrochemical biosensors, which may be roughly classed as affinity or catalytic biosensors. The former makes use of enzymes' trans-

ferability and catalysis. However, the necessity for enzyme activity to be maintained in a neutral environment has resulted in increasingly severe criteria for operating conditions and material selection; also, enzymes are rather expensive. Thus, creating more effective ways for designing electrochemical enzyme-free sensors, stabilizing immobilized media, and using and discovering efficient electron transfer media are only a few of the issues that need development in this sector.

Electrochemical sensors, in comparison to other sensing systems like calorimetric, magnetic, weight, piezoelectric, optical, and acoustic approaches, have a high sensitivity, are compatible with microfabrication technology, and are portable, economical, and simple to use. As a result, they are often employed in therapeutic settings. Neurobiological indicators are present at very low concentrations in biological fluids, necessitating the use of extremely sensitive detection technologies. Numerous ways have been used to change the electrode surface of (bio)sensors in order to increase their accuracy, sensitivity, and selectivity. Lyons and Clark introduced the notion of the glucose enzyme electrode for the first time in 1962; the field of biosensors has advanced significantly since then. Biosensors may be traced back to the enzyme electrodes that sparked their development. Due to the specificity of enzymes, electrochemical sensors based on enzymes demonstrate a high degree of selectivity and may therefore be used to measure individual enzymes in complex mixtures directly.

However, changes in environmental parameters such as pH and temperature have a significant effect on the activity of enzymes and other physiologically active chemicals, severely restricting the application range of enzyme biosensors. Additionally, the manufacturing procedure for the enzyme sensor is complex, and an uneven thickness might result in low repeatability and interference from oxygen. Furthermore, enzyme-based biosensors may be prohibitively expensive owing to the high cost of enzymes. As a result, the use and development of non-enzyme biosensors have become a new area of study. There are various enzyme-free methods for detecting biomarkers of neurodegenerative illnesses, including colorimetric sensors, surface-enhanced Raman scattering biosensors, fluorescence imaging sensors, and electrochemistry. Khalilzadeh et al. devised an electrochemical approach based on microRNA (miR) to detect miR-146a, a well-characterized biomarker for neurological illness. The capture microRNA (C-miR) was self-assembled on the gold surface and utilized to quantify the target microRNA (T-miR) of miR-146a in this bioassay. To do this, they immobilized an optimum concentration of C-miR on the surface of a gold electrode and utilized it to collect the target analyte (T-miR). The results obtained using an unprocessed human blood sample demonstrate unequivocally that the designed microRNA-based biosensor is capable of detecting miR-

146a as a biomarker for NDDs. Using Exo III-assisted recycling amplification and a graphene-modified electrode, Liu et al. created a ratiometric electrochemical biosensor for quantitative detection of the trinucleotide repeat sequence d(CAG)_n. The double-signals are hairpin DNAs tagged with ferrocene and methylene blue, which may hybridize to target DNA. This innovative ratiometric electrochemical biosensor offers an effective and reliable approach for analyzing d(CAG)_n trinucleotide repeats and may be used as a simplified clinical tool for neuro-degenerative illnesses.[6,14,21,24]

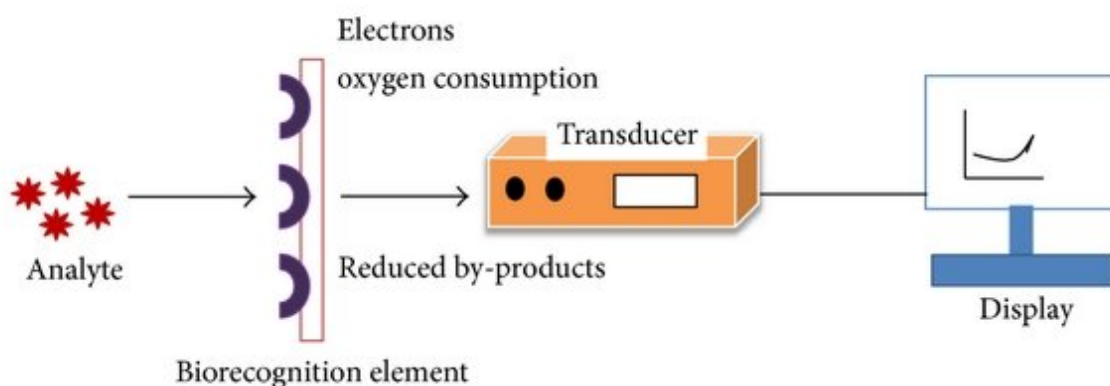


Figure 9: Electrochemical biosensor

Source: <https://www.creative-biolabs.com/drug-discovery/diagnostics/development-of-electrochemical-biosensor.htm>

Chapter 6

Challenges And Future Perspective:

Biosensors for neuro-degenerative disorders are also confronted with several challenges in their development and use. The nature of the disorders and the necessity of early, precise diagnosis make it very challenging to design effective biosensors. One of the main challenges is the identification and validation of unique biomarkers for various neuro- degenerative disorders. The best inflammatory biomarker should be easy to measure, reproducible, not liable to large variability within the population, and independent of external influence. Furthermore, the unavailability of biopsies for gold standard diagnosis makes it challenging to validate clinical diagnoses with pathological outcomes. Notably, new studies have investigated new methods of addressing these challenges. Salivary analysis, for example, has become more popular as a non-invasive procedure for the detection of biomarkers in neuro- degenerative disorders. Utilization of advanced materials like graphene derivative materials (GDMs) and metal-organic frameworks (MOFs) has proved useful in the fabrication of more sensitive and accurate biosensors. Yet, the incorporation of these new materials and methods into functional, point-of-care devices is still a challenge. Biosensors hold great promise for the early detection and monitoring of neuro- degenerative diseases, their evolution is challenged by biomarker discovery, sensitivity, specificity, and ready-to-use application. Overcoming these difficulties will necessitate ongoing multidisciplinary research and development in materials science, bio-engineering, and clinical neurology.

In the future, wearable, flexible, transportable conductive hydrogel-based, multifunctional, and miniature sensors for real-time individualized health monitoring and control of cybernetic prosthesis may be regarded as a rising branch of commercialized nano-biosensor applications. Exploiting innovative nanostructured materials such as borophene, photonic crystals, and quantum dots, as well as biological markers such as transcription factors, is only the tip of the iceberg in terms of potential. Additionally, nanotechnology advancements in biosensor development are revolutionary and could be accelerated further through strategic investments in smart technologies such as big data analytics, artificial intelligence (AI) with deep learning accessibility, and the Internet of things (IoT), which is built on advanced telecommunication networks. Emerging debates over nanobiosensors and smart technologies such as the Internet of Things (IoT) have spawned unique ideas such as the Internet of Bio NanoThings (IoBNT) and the Internet of NanoThings (IoNT). In basic words, the Internet of Things refers to any physical items, such as smartphones, that are linked to the internet and capable of exchanging data instantly through a unique identifier, allowing safe data transmission without the need for human-to-computer or human-to- human contact. Hybridizing this technology with the use of nanotechnology to detect biological macro-molecules with the accuracy and precision enabled by IoBNT is currently

being viewed as the future major intervention that warrants more investigation. These massive heterogeneous bigdata sets or data sets created may be rapidly turned into information with a high throughput value using AI with deep learning accessible, which would be prohibitively expensive and time-consuming otherwise. Combining AI, which is the emulation of human intelligence in robots meant to behave and think like humans, might be automated and upgraded. Combining AI-based biosensors for point-of-care detection with biological monitoring is the other developing area of modern hybridization. This combination could shed light on the critical role of algorithms of machine-learning in the development of futuristic nano- based biosensors, as well as microchip-based essential disease biomarkers, computational techniques, and the Internet of Things for patient compliance and real-time health monitoring. Nonetheless, nanomaterial-based biosensors have a lot of potential and will be widely used in environmental monitoring, process control, food analysis, and clinical diagnosis in the near future.[6-24]

Conclusions:

Biosensors provide a potentially transformative strategy for the precise and effective diagnosis of biological protein markers linked with a variety of communicable and non-communicable illnesses. The accuracy and precision may be enhanced by using nanotechnology, paving the way for nanobiosensors. The study provides a comprehensive overview of the use of nano-based biosensors in neurodegenerative illnesses, emphasizing many potential indicators. Because these intermediary molecules are regarded to be unique to the route in which they are engaged, designing a tailored biosensor for their detection may be advantageous. Additionally, as nanotechnology advances, innovative nano-structured materials such as borophene, photonic crystals, and quantum dots are emerging that may improve the persistence and accuracy of nanobiosensors. Biosensors, especially MOF-based biosensors, hold high promise for the early diagnosis and disease control of neurodegenerative disorders. Nevertheless, there are challenges regarding their clinical use and compatibility with current diagnostic tools. Development on these biosensors should aim to fine-tune them and couple them with other biomarker methods to enhance the timeliness and accuracy of neurodegenerative disease diagnosis and monitoring. The economic potential of nano-based biosensors is highlighted since it is one of the critical components necessary for enough personnel and financing. Future implementations or applications and conceptualizations of revolutionary computing methods, AI with access to deep learning, big data analytics, IoT, and microchip-based approaches coupled with nano biosensors might be a fantastic method for illness identification. As a result, research into these intelligent systems coupled with nanobiosensors must be promoted in order to advance the development of next-generation diagnostics. In our opinion, nanotechnology approaches are potent candidates for early detection of neurodegenerative disorders, since the smaller the transducer, the higher sensibility and selectivity and smaller amount of biomarkers are detectable.[6,14,21,22,23,24]

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symptoms, such as the se-
lective loss of brain cells, inflammation in the nervous system, the formation
of abnormal pro-
tein clumps, and damage to neurons and their connections. These issues
lead to significant
changes in thinking and behavior. Conditions like Alzheimer's disease (AD),
Parkinson's dis-
ease (PD), amyotrophic lateral sclerosis (ALS), frontotemporal dementia,
Huntington's disease
(HD), and prion diseases are all types of neurodegenerative disorders that
affect a large num-
ber of people globally. Among these, PD and AD are the most common
neurological diseases.
As people get older, the risk of developing a neurodegenerative disorder
increases significantly.
Recent studies suggest that a large number of people will be affected by
neurodegenerative con-
ditions in the coming decades, highlighting the urgent need to understand
the causes and de-
velop new methods for managing and treating these disorders. Research
estimates that approxi-
mately 30 million people around the world currently live with Alzheimer's
disease. The health,
economic and social impacts of neurodegenerative disorders are

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Aghili et al. [58], on the other hand, created an
electrochemical nanobiosensor to detect PD early using
the measurement of a circulating biomarker, miR-195.
Gold nanowires (CNWs) and exfoliated graphene oxide
were used to enhance the surface of a screen-printed
carbon electrode (EGO).

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